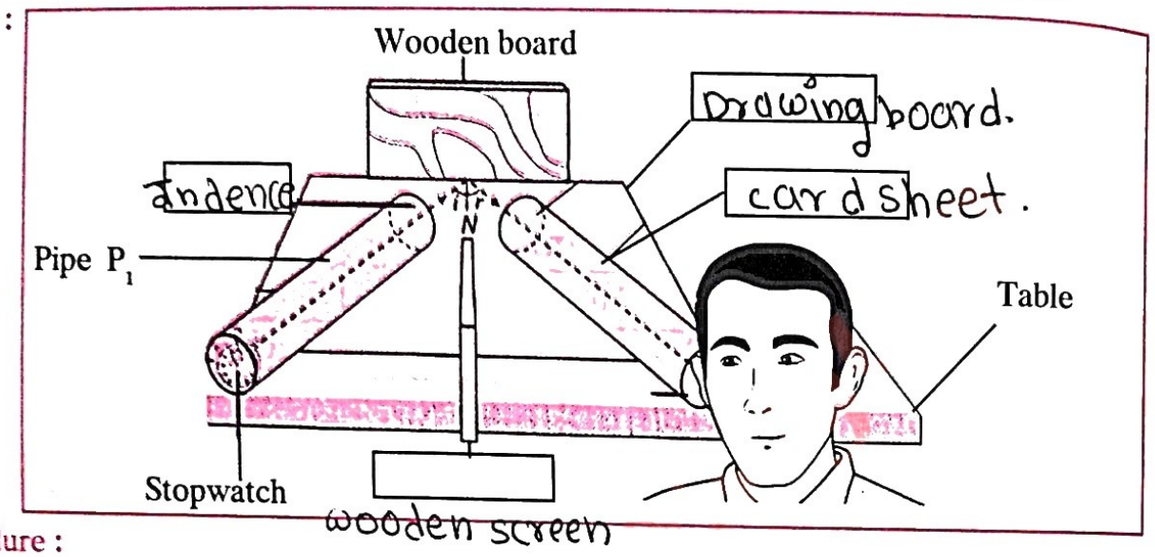
Experiment No. 10

Title : To verify the law of reflection of sound.

Apparatus : Drawing board, White paper, Stop watch, Mirror or Smooth reflecting surface such as metal plate, Cardsheet for making hollow tubes, Pins, Wodden screen etc.

Figure :



Procedure :

1. Arrange the apparatus as shown in figure.
2. Place a stopwatch at the end of the pipe P_1 as shown in figure.
3. Arrange the pipe P_1 so that the angle of incidence i = 30^0
4. Now place the pipe P2 in such a way that its one end is pointing in the direction of reflecting pipe.
5. Now take your ear near the other end of the pipe P_2 shown in figure and take the pipe P2 to a plane , where the ticking of the watch is heard clearly.
6. Measure the angle of reflection (r) and note it.
7. Repeat the above procedure for 40^0, 50^0, 60^0, 70^0 as the values of i .
8. For any one reading while the ticking of the watch is heard clearly lift the pipeP_2 slightly above the table in the same position and try to listen to the ticking of the watch.

Observation table :

Observation	Angle of incidence of sound (i)	Angle of reflection of sound (r)	Do you get i = r
1	30^0	30^0	Yes.
2	40^0	40^0	Yes.
3	50^0	50^0	Yes.
4	60^0	60^0	Yes.
5	70^0	70^0	Yes.
6	80^0	80^0	Yes.
7	90^0	90^0	Yes.

Inference :

1. The incident sound wave, reflected sound wave and normal lie in the same plane.
2. Angle of incidence = ...~~50°~~.....Angle of reflection.

Precaution :

1. The sound of the stopwatch should not reach to the listener directly.
2. Ear should be held close to the P_2.
3. The angle of reflection measure carefully where the reflected sound is the loudest.
4. The centermost part of the tubes should match perfectly with the line of incidence and the line of reflection.

Multiple Choice Questions

1. The audible sound range for human is .....
   A. 0-20Hz ✓B. 20Hz-20000Hz C. 20Hz-2000 Hz D. 100Hz-10,000Hz
2. When a sound wave transmitted through a medium, its .....is transported from one place to another.
   A. mass B. energy ✓C. velocity D. density
3. To make the the experiment for verifying the law of reflection of sound successfully .....should be used as an reflector.
   A. Sponge podded board. B. Clean white cloth
   ✓C. Perforated wooden board D. Polished wooden board

Questions

1. Explain the reasons behind the precautions to be taken in the experiment of reflection of sound.
   .....make two sufficient long any tubet out of a cardboard place a clock near the end the on tube and try hour the ticking of clock at the end of the other tube placing there adjust the angle of the two tubes till your can her the sound very clearly.....
2. Explain the advantages and disadvantages of the reflection of sound in everyday life.
   .Like light waves sound waves two get reflected from a solid of liquid sure face the sea waves also falls the laws of reflection a smather or a rough surface.....
3. Which remedies should you use to reduce the echo in auditorium?
   .An each is the reflection of the orginal sound becuse of reflection by same surface there for remedies you use to reduce the echo is auditorium.....

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